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# Supplementary material to RSHash

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## Abstract

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## 1. Bitvectors and MPHFs

We benchmarked rank operations for random values on the Elias-Fano-compressed bitvector $R$ (implemented by cite vigna) marking the existance of minimizers of length $17$ with maximum frequency $64$ on all datasets. Table 1 lists the running times for a rank operation and the computation of the Minimum Perfect Hash Function (MPHF) PTHash todo cite used in SSHash. Note that two rank operations $rank_R(m+1)$ and $rank_R(m)$ tell whether a Minimizer $m$ exists in the input (if they differ by one) and where to look in the size bitvector $S$. That corresponds to the value of a MPHF apart from that a MPHF cannot answer if a minmizer exists. For PTHash we used the standard parameters used in SSHash ...

## 2. Multiple Minimizer scheme

In addition to the effect of our multiple minimizer scheme on the space consumption we report its influence on streaming lookup times for low and high hitrates in Table 2. An additional level slows negative queries $27$ to $41$ ns per $k$-mer down, i.e., on average $40\%$, as negative queries require the computation of $l$ minimizers and $l$ bitvector lookups for $l$ levels. Positive queries become $13$ to $29$ ns slower, averaging a $20\%$ increase.

## 3. Parameter sensitivity

In Figure 1 we show the space consumption and negative streaming query times for all parameter configurations and each dataset. Note that for the datasets KESTREL and BACTERIAL that do not contain many repetitive regions adding more levels increases running time but does not reduce space consumption – here one level is the best choice. For the datasets HUMAN and COD the use of multiple minimizers reduces the required space notably while the running time increases as expected. Hence, the parameter choice is determined by the input sequences' repetitiveness and the desired time-space-tradeoff that is most sensitive to the number of minimizers and the minimizer length on the first level.

**Table 1.** Space requirement in bits per entry of a Elias-Fano compressed bitvector and a MPHF with the running times in ns for a rank operation and the computation of the hashfunction for minimizers of length $17$ that appear at most $64$ times in the datasets.

| | COD | | KESTREL | | HUMAN | | BACTERIAL | |
|---|---|---|---|---|---|---|---|---|
| | bits | ns | bits | ns | bits | ns | bits | ns |
| EF BV | 11.0 | 48 | 9.79 | 66 | 9.05 | 87 | 8.06 | 108 |
| MPHF | | | | | | | | |

**Table 2.** Space requirement in bits per $k$-mer (bpk) and $k$-mer streaming lookup times in ns per $k$-mer for our multiple minimizer scheme with up to three levels and different thresholds $t$ on the preprocessed human genome. Query times for low hitrates are denoted by $-$, queries with high hitrates by $+$. The minimizer lengths $m_i$ for levels $i \in [1,3]$ are $m_1 = 17$, $m_2 = 19$, and $m_3 = 21$.

| $t$ | 128 | | | 64 | | | 32 | | |
|---|---|---|---|---|---|---|---|---|---|
| Level | bpk | $+$ | $-$ | bpk | $+$ | $-$ | bpk | $+$ | $-$ |
| 1 | 10.7 | 83 | 78 | 14.1 | 71 | 73 | 14.0 | 62 | 74 |
| 2 | 9.21 | 112 | 107 | 9.21 | 90 | 114 | 11.0 | 75 | 111 |
| 3 | 8.02 | 135 | 134 | 8.50 | 104 | 145 | 9.43 | 88 | 151 |

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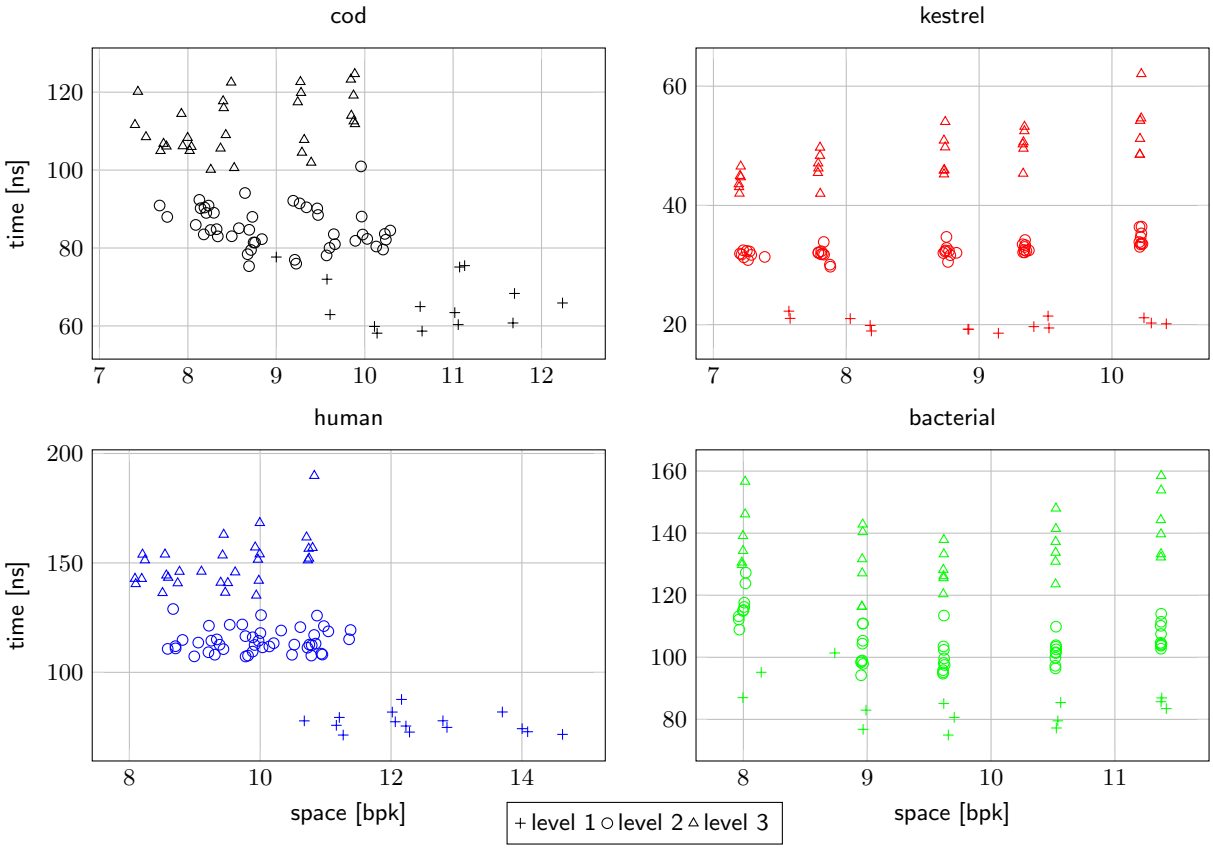



**Figure 1** Negative streaming query times and space requirement of RSHash for all parameter choices and for each dataset.